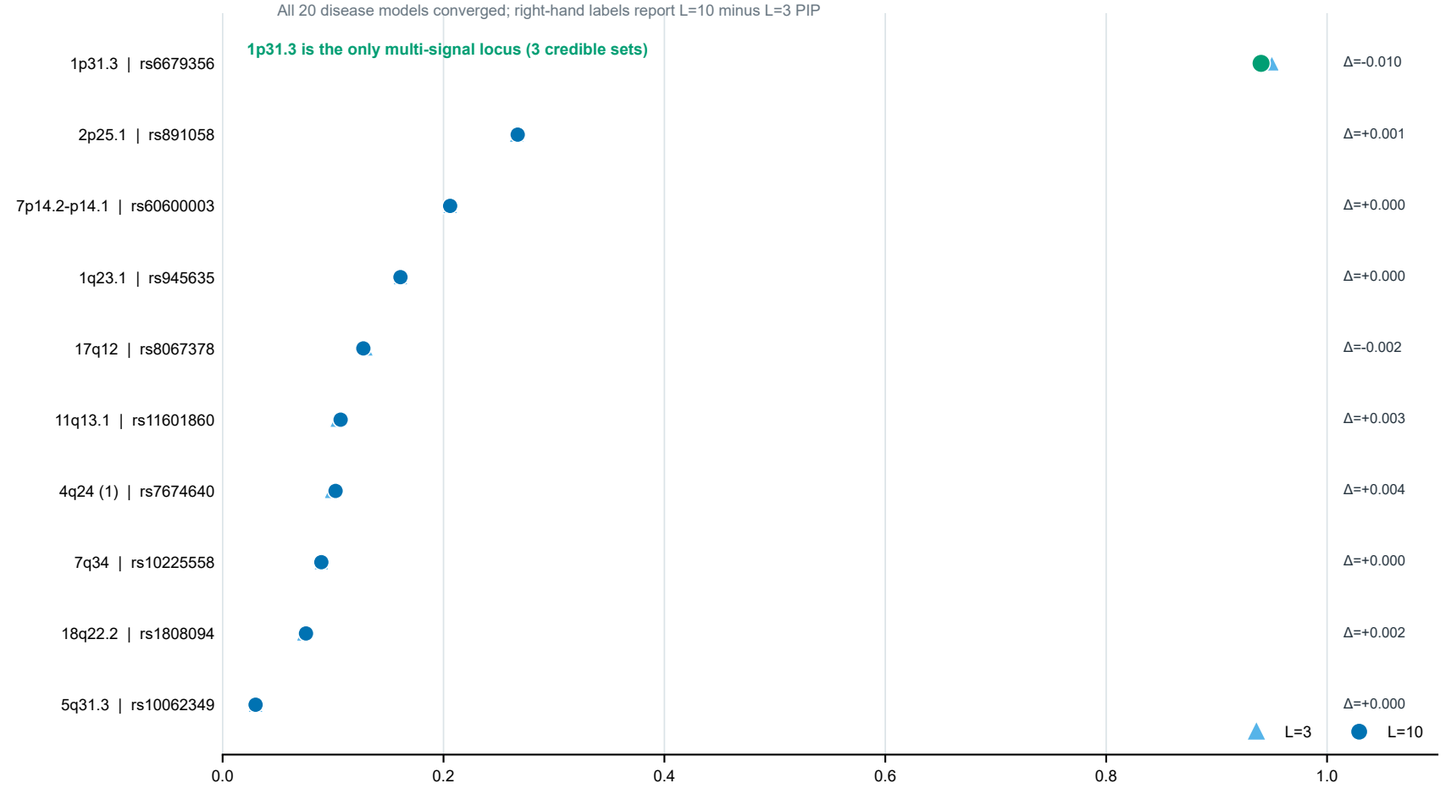
Local signed-LD and SuSiE quality control across 10 PBC loci

The exact 1000 Genomes EUR dosage-correlation matrices and variant order used for fine-mapping are audited

A Fine-mapping stability across model complexity

All 20 disease models converged; right-hand labels report L=10 minus L=3 PIP



B Exact signed-LD matrix audit

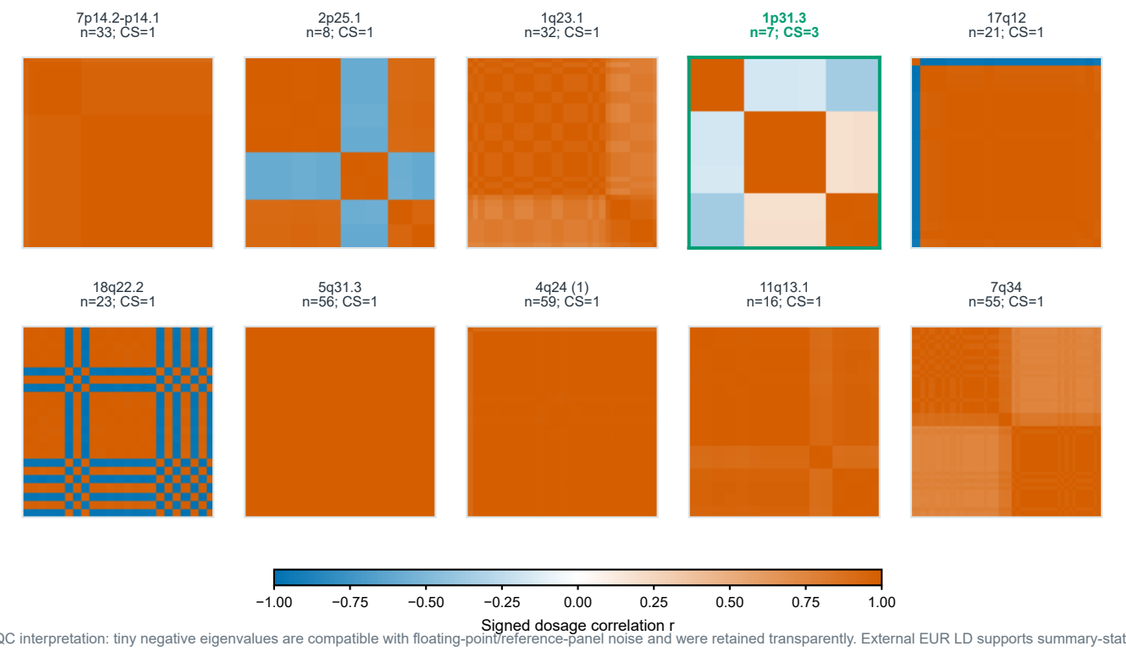
Full matrices are large; the audit reads every diagonal element and each credible-set block

Matrix byte size	10 / 10 exact	Finite CS-block entries	10 / 10
PASS		PASS	
Diagonal error <= 1e-6	10 / 10	Symmetry error <= 1e-6	10 / 10
PASS		PASS	
Exact variant order	10 / 10	Allele alignment	10 / 10 ready
PASS		PASS	

Reference LD: 1000 Genomes EUR dosage correlation; external to the discovery GWAS.

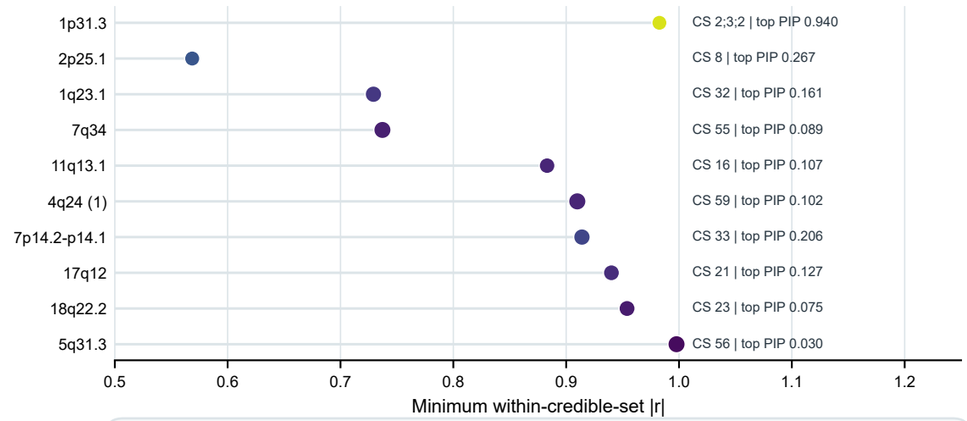
C Signed-LD within the L=10 credible sets

Each tile uses the exact credible-set union; blue denotes negative signed r, not failed LD



D Credible-set resolution and RSS diagnostics

High purity does not imply a small credible set or a causal gene



RSS s-hat range: 0.027–0.083 Numerically negative eigenvalues: 13 total; min -2.14e-06

2p25.1 has the weakest CS purity (0.569); broad high-LD sets at 5q31.3/4q24/7q34 remain poorly resolved.

QC interpretation: tiny negative eigenvalues are compatible with floating-point/reference-panel noise and were retained transparently. External EUR LD supports summary-statistic fine-mapping but does not replace study-specific LD or individual-level conditional analysis.